

Verifier Settings

1. Model Configuration

The PAIFAR verifier uses GPT-4o, the same model used for intervention-necessity judgment, but performs a distinct task: evaluating the quality and constraint satisfaction of a generated rewrite.

Model Configuration

Model: GPT-4o Temperature: 0 Maximum output tokens: 512 Request timeout: 120 seconds Retries: 3 Output format: JSON
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2. Evaluation Dimensions

2.1. For LOCAL_REPAIR, the verifier evaluates:

Debiasing: whether the original gender bias has been appropriately removed without introducing new gender bias.

Naturalness: whether the rewritten Chinese is fluent, complete, and natural.

Fidelity: whether the core non-bias-related content and communicative function of the original are preserved.

2.2. For PROPOSITION_RECONSTRUCTION, fidelity is replaced by:

Relevance: whether the reconstructed sentence appropriately addresses and repairs the core biased proposition while preserving the main topic or key meaning.

Each dimension receives a score from 1 to 3.

2.3. LOCAL_REPAIR vs. PROPOSITION_RECONSTRUCTION

Core distinction:

The distinction is not how much is changed, but whether the gender bias is attached to the proposition or constitutes the proposition itself.

LOCAL_REPAIR → Keep the proposition, remove the bias.

Example: “Although she is a woman, she is well suited to be a leader.”

- Keep: She is well suited to be a leader.
- Evaluation: Fidelity

PROPOSITION_RECONSTRUCTION → The proposition itself is biased, so reconstruct it.

Example: “Women are naturally unsuitable for leadership.”

- No non-biased proposition to preserve.
- Evaluation: Relevance

In short:

Local → Preserve → Fidelity

Reconstruction → Reconstruct → Relevance

3. Quality Score

The final quality score is computed programmatically from the verifier's dimension scores:

Debiasing: 50% · Naturalness: 25% · Type-specific criterion: 25%

Score design principle: Debiasing is the core, Naturalness is the foundation, and the type-specific criterion ensures either Fidelity or Relevance.

The three 1–3 scores are normalized to a 0–100 scale before weighted aggregation.

4. Acceptance and Refinement

A candidate is accepted when its weighted quality score reaches the predefined threshold of 80/100.

Candidates that do not meet the acceptance condition are passed to the feedback-guided refinement process. The maximum number of refinement rounds is 3. If no candidate satisfies the acceptance condition, the system selects the highest-ranked candidate according to the predefined selection policy.

Additionally, threshold and maximum number can be adapted according to different scenarios